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The Impact of Localized Muscle Mass on Sprint Mechanics During the Swing and Stance Phases of Division I Collegiate American Football Players

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Abstract

Metoyer, CJ, Lever, JR, McGinty, M, Landow, L, Hunt, R, Hauenstein, JD, Huebner, A, Stone, MH, and Wagle, JP. The impact of localized muscle mass on sprint mechanics during the swing and stance phases of Division I collegiate American football players. *J Strength Cond Res* XX(X): 000–000, 2025—This study evaluates the impact of localized muscle volume on sprint mechanics among Division-I collegiate American football players across 3 position groups: Skills (Defensive Backs, Wide Receivers), Mids (Linebackers, Running Backs, Tight Ends), and Bigs (Offensive, Defensive Linemen). A total of 108 male athletes (age: 21.4 ± 1.9 years, BMI: 29.6 ± 4.3) were assessed. Lower extremity muscle volumes were measured using magnetic resonance imaging, and sprint kinematics were captured using an inertial motor unit system during maximal effort sprints. Stepwise multiple linear regressions were conducted to evaluate the relationship between muscle volumes, stance, and swing times for each position group. In the Skills group, increased volumes in the vastus intermedius ($\beta = -0.15$, $p < 0.01$) and sartorius ($\beta = -0.17$, $p = 0.01$) were associated with reduced swing time, while those in the vastus medialis ($\beta = 0.18$, $p < 0.01$) increased swing time. For stance time, the gluteus maximus ($\beta = -0.02$, $p = 0.03$) was associated with reduced stance time, and the tibialis posterior ($\beta = 0.28$, $p < 0.01$) increased it. In the Mids group, the biceps femoris long head ($\beta = -0.15$, $p < 0.01$) was associated with reduced stance time, while the vastus medialis ($\beta = 0.09$, $p < 0.01$) increased it. In the Bigs group, the rectus femoris and sartorius were associated with reduced stance time, while the tensor fasciae latae and tibialis anterior increased stance time. These findings highlight the role of localized muscle volume in influencing sprint mechanics and emphasize the potential need for position-specific physical development programs tailored to the biomechanical demands of running mechanics in American football players.

Key Words: muscle volume, hypertrophy, running, sprint kinematics, positional, MRI, inertial motor unit

Introduction

Running speed can be a decisive factor in competitive outcomes in American football, affecting offense, defense, and special teams (3,14). Thus, understanding the contributing factors to maximizing sprint performance has garnered considerable interest among sport researchers and practitioners. Optimizing sprint mechanics, such as proper stride length, swing time, and ground contact time, is essential for maximizing propulsion and minimizing energy expenditure (6), thereby enabling athletes to achieve higher speeds with improved economy of motion (1). Although running form is an emergent outcome of the interactions of many factors, the regional distribution of muscle volume on the lower extremity logically plays an important role, because it influences force production, joint mechanics, and movement efficiency during each phase of the gait cycle. For example, more muscle volume distally (e.g., around the knee) would increase the torque required to lift the leg during swing, thus potentially positioning the limb at a mechanical disadvantage entering the subsequent foot strike (13). However, adequate muscle volume is instrumental in generating the propulsive forces needed for rapid acceleration. Therefore, achieving the right distribution of muscle volume in the appropriate regions is crucial for developing speed in football athletes.

Previous studies have extensively examined overall muscle size in the context of sprinting, reporting positive correlations with sprint performance. For instance, Kubo et al. (8) demonstrated that increased quadriceps cross-sectional area is associated with improved sprint times. Similarly, Mangine et al. (9) found that muscle thickness in the quadriceps significantly predicts sprint performance. However, it has also been reported that generalized hypertrophy can lead to decreased flexibility and nonfunctional mass, leading to altered sprint kinematics and a decrease in performance. For example, Trappe et al. (15) indicated that excessive muscle size might impair movement efficiency and flexibility.

Although research shows mixed results with generalized muscle size, the nuances of regional muscle volume distribution and its potential impact on the kinematics and kinetics characterizing sprinting's swing and stance phases are less understood. Traditionally, research has predominantly focused on muscle volume at the lower extremity level overall, often overlooking potential region- or muscle-specific adaptations within the lower extremities. Therefore, the primary aim of this study is to investigate the influence of localized muscle volume within the lower extremity musculature, on sprint mechanics in American

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football players. This study primarily deepens our understanding of the intricate biomechanics underlying sprinting performance, while its cross-sectional findings offer valuable insights that may guide future research, and the development of targeted training interventions aimed at improving sprint speed.

Methods

Experimental Approach to the Problem

This study analyzed existing data from the larger Hamstring Injury (HAMIR) study project, a multisite prospective investigation into the etiology of hamstring strain injuries in American football players (7). Subsequently, a cross-sectional study design was adopted to evaluate the relationship between localized lower-extremity muscle volumes and sprint mechanics in Division-I collegiate football athletes across 3 positional groups. These groupings were based on commonly recognized cohorts, termed Skills (Defensive Backs and Wide Receivers), Mids (Linebackers, Running Backs, and Tight Ends), and Bigs (Offensive and Defensive Lineman), as per the HAMIR project (7). Muscle volumes were assessed using magnetic resonance imaging (MRI) scans enhanced through machine learning, capturing detailed structural data for each athlete's lower extremities. Sprint kinematics were recorded using a sensor-based inertial motor unit (IMU) system during overground maximal effort sprint trials.

Subjects

Data from 108 male NCAA Division I American football players (age: 21.5 ± 1.9, BMI: 29.2 ± 4.5) were analyzed for this study (Table 1). Ethical approval for the study was obtained from the Health Sciences Institutional Review Boards at the University of Wisconsin-Madison and at each participating data collection site (23-02-7689). All participants provided written informed consent. The Collaborative Research Network of the American Medical Society for Sports Medicine serves as the Data Coordinating Center for this study.

Procedures

Sprint Kinematics. Sprint kinematics were captured using a sensor-based IMU system (MTw Awinda; Xsens Technologies B.V., The Netherlands). Seven IMUs were affixed to each subject using adhesive tape and/or customized straps at the sacrum, left and right thigh, left and right shank, and left and right foot. Calibration of the IMUs was conducted using the proprietary manufacturer software (Xsens MVN, Xsens Technologies B.V.). After calibration, a static trial was conducted with subjects standing in the anatomical position to provide baseline data for sensor orientation.

After completing the static trial, subjects performed 2 maximal velocity over-ground sprinting trials from a standing start of approximately 30 meters each, with a minimum of 2 minutes of rest between each trial to minimize fatigue effects. Kinematic data were collected across all phases of the sprint, including acceleration, maximal velocity, and deceleration.

Table 1
Subject descriptive statistics.

	Skills (n = 43)	Mids (n = 28)	Bigs (n = 37)
Age (yr)	21.5 ± 1.9	21.5 ± 1.9	21.6 ± 2.0
BMI (kg·m^{-2})	29.31 ± 4.59	29.34 ± 4.57	29.36 ± 4.70
Swing time (ms)	313.31 ± 25.64	313.20 ± 25.61	314.72 ± 25.48
Stance time (ms)	156.14 ± 21.83	155.88 ± 21.84	156.48 ± 22.00

Muscle Volume Assessment. Lower extremity muscle volumes were assessed using MRI, conducted using a Signa Pioneer 3T scanner (General Electric, Milwaukee, WI) with a flexible surface coil. The scans covered the lumbopelvic region and both lower extremities to assess muscle volumes across the entire field of view (FOV). Lower extremity imaging used axial multislice 2D gradient echo fast SPGR sequences, optimized for high-muscle signal and low-fat signal. Key imaging parameters included TR/TE of 839 ms/4.35 ms, a FOV of 500 × 344 mm, a 256 × 176 matrix, and a voxel size of 2 × 2 × 5 mm with fat suppression. Contiguous axial images were acquired from the 12th thoracic vertebra to below the ankle joint in sets of 40 images (7).

Analysis of the MR images included quantification of all viewable muscle volumes within the target area. Automatic muscle segmentation was performed using Springbok machine learning technology (Springbok, Charlottesville, VA), with algorithms updated and refined to include automatic segmentation of tendons, scar, and edema based on manually segmented training datasets (12).

Statistical Analyses

A stepwise multiple linear regression analysis was performed to identify the most relevant predictors (i.e., muscle volumes) of swing and stance time across different position groups in American Football. The Akaike Information Criterion (AIC) was used as the criterion for model selection, optimizing the balance between model fit and complexity to determine the best-performing model. The stepwise procedure implemented using the stepAIC function in R (MASS package version 7.3–60, R Core Team, Vienna, Austria) iteratively added or removed variables to minimize the AIC value. Although the primary goal was to identify the optimal model, the selected predictors inherently represent the variables that contributed most significantly to the model's explanatory power.

Residual diagnostics confirmed that the final models met the assumptions of linearity, homoscedasticity, and normality. The results are presented as regression coefficients (β), confidence intervals, and p-values for each predictor variable. In addition, standard statistical procedures were used to calculate means and *SD*. All data are presented as mean ± *SD* unless otherwise indicated (M ± *SD*). Statistical significance was set at $p \leq 0.05$ for all tests.

Results

Table 2 presents the means and *SD*s of muscle volumes and their relationship with swing and stance time for the entire sample. The strongest positive correlations observed between both swing and stance time with muscle volumes were for both the vastus medialis and Soleus muscles (r = 0.33, 0.33, 0.49, and 0.48, respectively).

Multiple linear regressions were performed using Stepwise AIC to assess the association between muscle volumes and both swing (Table 3) and stance time (Table 4) for each position group. The final models displayed statistical significance for each position group, highlighting the influence of positional demands on specific muscles and their effect on sprint mechanics.

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Table 2
Muscle volume means and correlations.

	Mean ± *SD*	Swing time *r*	Stance time *r*
Swing time		—	0.33
Stance time		0.33	—
Vastus medialis	811.64 ± 153.81	0.33	0.49
Vastus intermedius	455.170 ± 96.91	0.28	0.43
Vastus lateralis	1,524.48 ± 257.91	0.31	0.47
Tibialis posterior	127.28 ± 27.20	0.24	0.31
Tibialis anterior	166.19 ± 35.84	0.21	0.37
Tensor fascia latae	124.42 ± 36.12	0.12	0.20
Soleus	609.78 ± 123.70	0.33	0.48
Semitendinosus	378.01 ± 104.43	0.1	0.08
Semimembranosus	389.59 ± 71.64	0.22	0.30
Sartorius	271.33 ± 57.95	0.06	0.32
Rectus femoris	441.65 ± 69.66	0.2	0.23
Psoas major	447.50 ± 66.16	0.08	0.18
Piriformis	58.47 ± 27.22	0.1	0.02
Gracilis	194.68 ± 38.74	0.1	0.31
Gluteus maximus	1,606.07 ± 276.05	0.17	0.36
Gluteus medius	472.26 ± 73	0.23	0.40
Gluteus minimus	133.30 ± 23.66	0.25	0.40
Gastroc (medial)	369.30 ± 87.32	0.23	0.33
Gastroc (lateral)	215.75 ± 51.54	0.26	0.38
Biceps femoris (short)	23.23 ± 6.15	0.25	0.32
Biceps femoris (long)	319.43 ± 67.84	0.14	0.25
Adductor magnus	1,105.48 ± 180.96	0.28	0.43
Adductor longus	280.48 ± 46.39	0.24	0.36

For the Skills group, the stance time model indicated that the gluteus maximus had a significant negative association with stance time ($\beta = -0.02$, $p = 0.03$), suggesting that greater muscle volume in this region reduces stance time. In contrast, the tibialis posterior demonstrated a significant positive association with stance time ($\beta = 0.28$, $p < 0.01$), indicating that increased muscle mass in this muscle prolongs stance time. In the swing time model, the sartorius ($\beta = -0.17$, $p = 0.01$) and vastus intermedius ($\beta = -0.15$, $p = 0.01$) were associated with shorter swing times, while the vastus medialis exhibited a significant positive association with swing time ($\beta = 0.18$, $p < 0.01$), indicating that hypertrophy in the vastus medialis may impede swing mechanics by increasing swing time.

Table 3
Stepwise Akaike Information Criterion regression results for swing time.

IV	β	Confidence intervals Lower	Confidence intervals Upper	*p*
Skills				
Biceps femoris long	0.11	−0.01	0.22	0.06
Biceps femoris short	0.2	−0.05	0.46	0.12
Glute medius	−0.11	−0.23	0.01	0.08
Rectus femoris	0.08	−0.02	0.19	0.11
Sartorius	−0.17	−0.3	−0.04	0.01
Soleus	0.08	−0.01	0.17	0.07
Vastus intermedius	−0.15	−0.26	−0.03	0.01
Vastus medialis	0.18	0.09	0.27	<0.01
Mids				
Biceps femoris long head	−0.2	−0.31	−0.1	<0.01
Biceps femoris short	0.27	0.07	0.46	<0.01
Glute maximus	−0.07	−0.1	−0.04	<0.01
Piriformis	−0.31	−0.48	−0.14	<0.01
Rectus femoris	−0.1	−0.19	0	0.05
Semimembranosus	0.23	0.1	0.36	<0.01
Bigs				
Piriformis	0.53	0.13	0.93	0.01
Psoas major	−0.1	−0.22	0.02	0.11
Sartorius	−0.11	−0.23	0.01	0.06
Semimembranosus	−0.14	−0.25	−0.03	0.01
Vastus lateralis	0.05	0.02	0.08	<0.01

Table 4
Stepwise Akaike Information Criterion regression results for stance time.

Independent variable	β	Confidence intervals Lower	Confidence intervals Upper	*p*
Skills				
Glute maximus	−0.02	−0.04	0	0.03
Tibialis posterior	0.28	0.15	0.42	<0.01
Mids				
Biceps femoris long head	−0.15	−0.22	−0.08	<0.01
Sartorius	0.06	−0.02	0.14	0.2
Semimembranosus	0.05	−0.02	0.13	0.2
Soleus	0.04	0	0.08	0.05
Vastus intermedius	−0.14	−0.23	−0.05	<0.01
Vastus medialis	0.09	0.04	0.14	<0.01
Bigs				
Adductor magnus	0.03	0	0.06	0.04
Biceps femoris short head	0.12	−0.03	0.27	0.11
Rectus femoris	−0.11	−0.18	−0.03	<0.01
Sartorius	−0.13	−0.24	−0.02	0.03
Tensor fasciae latae	0.33	0.17	0.5	<0.01
Tibialis anterior	0.2	0.02	0.38	0.03

In the Mids group, the stance time model revealed that the biceps femoris long head had a significant negative association with stance time ($\beta = -0.15$, $p < 0.01$), suggesting that increased muscle mass in this region shortens stance time. In addition, the vastus intermedius also showed a negative association ($\beta = -0.14$, $p < 0.01$), while the vastus medialis displayed a positive association with stance time ($\beta = 0.09$, $p < 0.01$). For swing time, the biceps femoris long head, piriformis, and semimembranosus ($\beta = -0.20$, -0.31, and 0.23, respectively) all showed significant associations ($p < 0.01$), indicating that while the biceps femoris long head and piriformis reduce swing time, hypertrophy in the semimembranosus increases swing time.

For the Bigs group, the stance time model showed significant associations ($p < 0.01$) for the rectus femoris, sartorius, and tensor fasciae latae (TFL) ($\beta = -0.11$, -0.13, and 0.33, respectively), indicating that increased volume in the rectus femoris and sartorius reduces stance time, while the TFL increases stance time. In addition, the tibialis anterior was also positively associated with stance time ($\beta = 0.20$, $p = 0.03$). In the swing time model, significant associations were found for the piriformis, semimembranosus, and vastus lateralis ($\beta = 0.53$, -0.14, 0.05, respectively; $p \leq 0.01$), indicating that the semimembranosus reduces swing time, while the piriformis and vastus lateralis increase it.

Discussion

The purpose of this study was to evaluate how individual lower-extremity muscle volumes influence sprint mechanics among collegiate American football athletes. The regression analyses revealed that variations in muscle size significantly affect the stance and swing phases of sprinting, with these relationships differing by position group.

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For the Skills group, greater gluteus maximus volume was associated with reduced stance time ($\beta = -0.02$, $p = 0.03$), likely reflecting its role in generating horizontal momentum during hip extension. A shorter stance time enhances the ability to store and return elastic energy, providing a key advantage for skill players who rely on speed to create or limit separation depending on their offensive or defensive roles. In contrast, increased tibialis posterior volume was associated with prolonged stance time ($\beta = 0.28$, $p < 0.01$), suggesting that larger distal muscle volumes may hinder optimal ground contact mechanics. These findings imply that training interventions for skill players might benefit from emphasizing proximal muscle development, such as through back squats, to enhance sprint efficiency.

Swing phase mechanics were also influenced by muscle volumes in the Skills group. Shorter swing times were associated with greater sartorius and vastus intermedius volumes ($\beta = -0.17$ and $\beta = -0.15$, respectively; $p = 0.01$), suggesting these muscles play a role in leg recovery during the swing phase. Although the sartorius contributes to hip flexion, the vastus intermedius likely aids indirectly by supporting knee extension and overall limb control. Conversely, greater vastus medialis volume was associated with increased swing time ($\beta = 0.18$, $p < 0.01$), potentially because of its distal position and the associated mechanical workload during leg lift. These findings further underscore the potential benefits of prioritizing proximal muscle hypertrophy over distal muscle development for skill players aiming to optimize sprint performance.

Prior research supports these findings by demonstrating that resistance training elicits distinct, localized changes in muscle volume, influenced by exercise selection. For example, Narici et al. (11) reported preferential hypertrophy of the vastus medialis after isokinetic knee extension training, likely because of its activation patterns. Similarly, Gullett et al. (5) showed that back squats emphasize hip extensors, while front squats preferentially target the quadriceps, with studies by Yavuz et al. (17) and Trindade et al. (16) confirming that front squats consistently elicit higher quadriceps activation across varying knee angles. These findings highlight the importance of exercise specificity in driving muscle volume adaptations, which in turn influence sprint mechanics.

In this study, average stance and swing times for the Skills group were 156.14 ± 21.83 ms and 313.31 ± 25.64 ms, respectively, closely aligned with ranges reported for elite sprinters (10). Although these values reflect the explosive demands of American football, they also underscore the sport-specific need to balance sprint mechanics with agility, power, and directional changes. By tailoring training programs to emphasize proximal muscle development while avoiding excessive distal hypertrophy, practitioners may better align resistance training with the biomechanical demands of the sport.

For the Mids group, larger biceps femoris long head and vastus intermedius volumes were significantly associated with decreased stance time ($\beta = -0.15$ and $\beta = -0.14$, $p < 0.01$), reflecting their role in enhancing leg turnover during ground contact. Conversely, greater vastus medialis volume continued to demonstrate hindrances in sprint mechanics, increasing stance time ($\beta = 0.09$, $p < 0.01$), likely because of the higher torque demands associated with this muscle. Regarding swing time, increased biceps femoris long head and piriformis volumes reduced duration ($\beta = -0.20$ and $\beta = -0.31$, $p < 0.01$), supporting efficient leg recovery, while greater semimembranosus volume prolonged swing time ($\beta = 0.23$, $p < 0.01$), potentially impeding recovery during the swing phase. These results suggest that, similar to skill players, Mids require targeted training to optimize proximal muscle development while managing vastus medialis hypertrophy, because this muscle seems to hinder sprint efficiency. These similarities are logical considering the overlap in competitive demands between these groups.

For the Bigs group, rectus femoris ($\beta = -0.11$, $p < 0.01$) and sartorius ($\beta = -0.13$, $p = 0.03$) volumes were associated with reduced stance time, suggesting their contribution to improving ground contact mechanics. However, larger TFL volume increased stance time ($\beta = 0.33$, $p < 0.01$), likely reflecting its role in stabilization during ground contact in larger-bodied athletes, albeit at the potential cost of sprinting efficiency. This is likely an overall advantage to these groups, however, because this stability is important for impact encounters with opposing players. For swing time, greater semimembranosus volume reduced duration ($\beta = -0.14$, $p = 0.01$), while increased piriformis and vastus lateralis volumes prolonged it ($\beta = 0.53$ and $\beta = 0.05$, $p < 0.01$). Interestingly, piriformis volume demonstrated contrasting effects between groups, reducing swing time in Mids but increasing it in Bigs, possibly reflecting differences in positional demands and the limited maximal sprinting typical of Bigs. These findings underscore the role of position-specific demands in influencing muscle–sprinting mechanics relationships. Although Mids balance speed and power for agility and physicality in blocking and tackling, Bigs prioritize stabilization and strength to meet their positional requirements.

Although these results provide valuable insights, this study has several limitations. First, data were collected across multiple sites, potentially introducing variability in procedures (e.g., warm-up protocols, time of day). However, the large sample size supports the ecological validity and generalizability of the findings. Second, although individual muscle volumes were analyzed, practical training applications typically target coordinated muscle groups rather than isolated muscles. Despite this limitation, analyzing individual muscle volumes provides crucial insights into their biomechanical contributions and aids in selecting appropriate assistance and accessory exercises in resistance training. In addition, an important limitation is the potential influence of muscle volume on change of direction performance. However, assessing change of direction was beyond the scope of this study, because only linear sprint kinematics were measured. Future research should investigate how muscle volume affects change of direction kinematics in American football athletes to address this gap in performance analysis. Finally, the study focused on muscle volume without considering rate of force development, which likely interacts with volume to influence sprint mechanics. Future research should examine the combined effects of muscle volume and force production to gain a deeper understanding of their roles in optimizing sprint performance, while also controlling for injury data to determine the influence on injury risk profiles.

In conclusion, these findings highlight the influence of localized muscle volumes on sprint biomechanics and emphasize the need for position-specific strength and conditioning programs. For Skills group athletes, increasing sartorius and vastus intermedius volumes while managing vastus medialis hypertrophy could optimize leg recovery and sprint efficiency. For Mids, targeting proximal muscle hypertrophy while addressing potential inefficiencies associated with the vastus medialis could balance speed and power. For Bigs, careful management of TFL and piriformis volume is key to maintaining stability without compromising sprint mechanics. These results demonstrate that muscle volume adaptations are highly position dependent, reinforcing

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the importance of tailoring training programs to the biomechanical demands of different roles to enhance sprint performance and overall athletic success.

Practical Applications

This study emphasizes the need for targeted strength and conditioning programs to optimize sprint mechanics in collegiate American football players. For Skill players, exercises that target the sartorius and vastus intermedius, such as lunges and hip flexor strengthening, can help reduce swing time and improve leg recovery. Conversely, careful programming is necessary to prevent excessive development of the vastus medialis, which may contribute to increased swing time. Although front squats preferentially target the quadriceps including the vastus medialis, their volume and intensity should be managed appropriately within a balanced training program to avoid excessive hypertrophy in this region. For Mids and Bigs, focusing on exercises that enhance the biceps femoris long head and rectus femoris, such as Nordic hamstring curls and Romanian deadlifts, may aid in reducing stance and swing times. In addition, incorporating plyometrics improves rate of force development, ensuring functional power output (4). Furthermore, this may aid in the transfer of training effect leading to improved on-field performance (2). Finally, incorporating this information into tailored, position-specific training strategies is essential to optimize sprint performance across different positional demands, ensuring the potential for high performance.

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